

Yi Gao

Robot Learning · Contact-Rich Manipulation · Humanoid Compliance

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Available immediately for remote research collaboration: 30 hours/week

Education

University of Delaware

Sep 2024 - Jun 2028 (expected)

Southwestern University of Finance and Economics

SWUFE-UD Institute of Data Science · China-based dual-degree program

B.S. in Information Systems · GPA: 87.7/100

Research Experience

SymBiosis Robotics

May 2026 - Present

Research Intern; ongoing remote collaboration · Advisor: Pengxiang Ding

SoftSONIC: Residual Learning for Humanoid Compliance

- Independently led SoftSONIC from method design and data generation to RL experiments and deployment validation, with research guidance from senior collaborators. Studied lightweight compliance adaptation of an existing humanoid motion prior.
- Developed a bounded **64-D latent residual** over the frozen SONIC encoder and decoder; trained it with PPO and SoftMimic-inspired compliant references. The deployed actor uses proprioceptive/action history without explicit external-force measurements.
- Trained and evaluated a **shared residual policy on 32 training motions in simulation**, using paired perturbations and no-force tests; compared joint-space residuals, nominal-token conditioning, and reward variants to study compliance, motion preservation, and transfer.
- Implemented and validated a **50 Hz C++/TensorRT deployment** based on official SONIC. Observed qualitative yielding under manual pulls on a **Unitree G1** during standing, walking, and selected motion replays, including a backward-walking reference excluded from residual-policy training. Observations remain preliminary; response strength varies across motions.

Lenovo Robotics Research Institute (Shanghai)

Jul 2025 - Dec 2025

Research Intern · Solution & Development Center

- Developed a context-grounded robot agent prototype for industrial patrol, integrating visual perception, location/time-specific safety rules, and action protocols into a traceable decision pipeline.
- Collected and organized **200+ real-world video clips** using a DEEPRobotics X30 quadruped; built a structured annotation and VLM evaluation workflow comparing QwenVL and GLM-family models.
- Drafted *CE4Patrol: Multi-Layer Context Reasoning for Industrial Anomaly Inspection* (manuscript; Yi Gao, et al.).

Selected Technical Project

VLA Inference for Retail Mobile Manipulation

Mar 2026 - Apr 2026

Reproduced a **RoboBenchMart** VLA inference pipeline in ManiSkill for retail mobile manipulation. Set up assets, scene generation, model serving, and evaluation-client integration for pick-and-place and open/close tasks.

Technical Skills

Programming & tools: Python, C++, PyTorch, Git, Linux, Docker, TensorRT

Simulation: Isaac Lab / Isaac Sim, MuJoCo, ManiSkill, PyBullet

Methods: PPO, residual policy learning, motion tracking, imitation learning, experimental diagnostics

Honors & Languages

Meritorious Winner, Mathematical Contest in Modeling (COMAP), 2025

Second Prize, National English Competition for College Students, 2025 · CET-4: 632; CET-6: 599